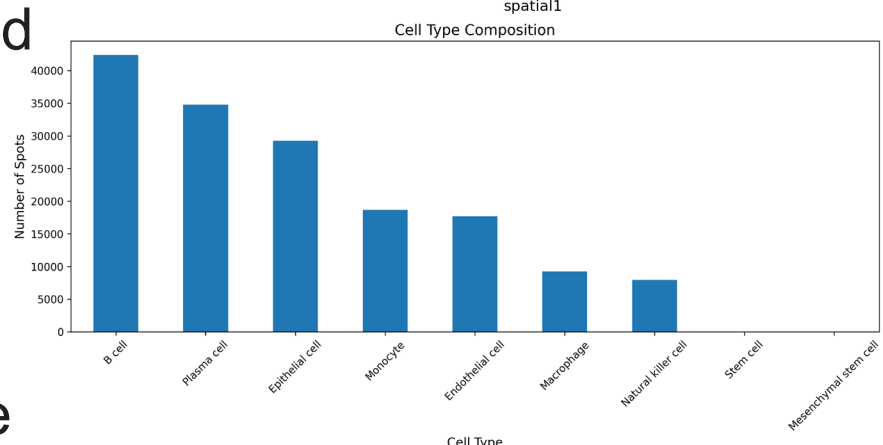
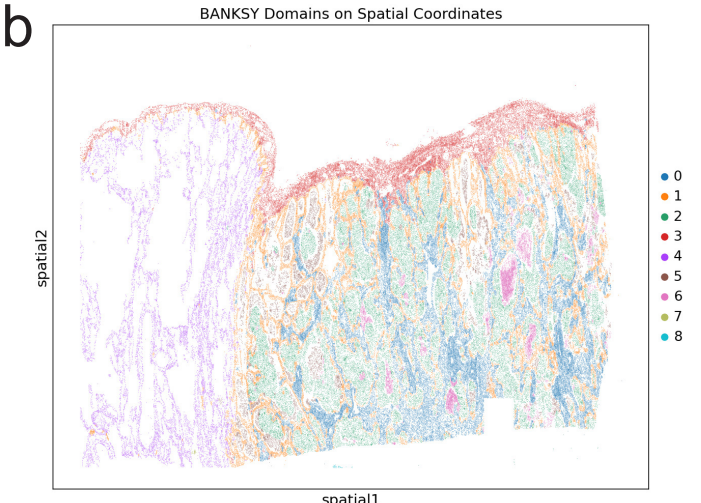
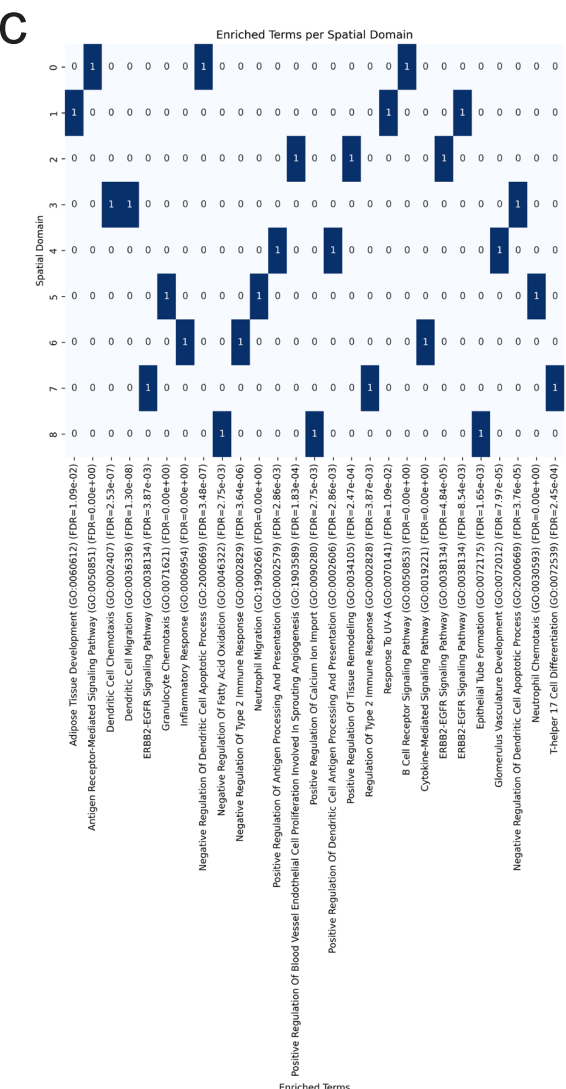
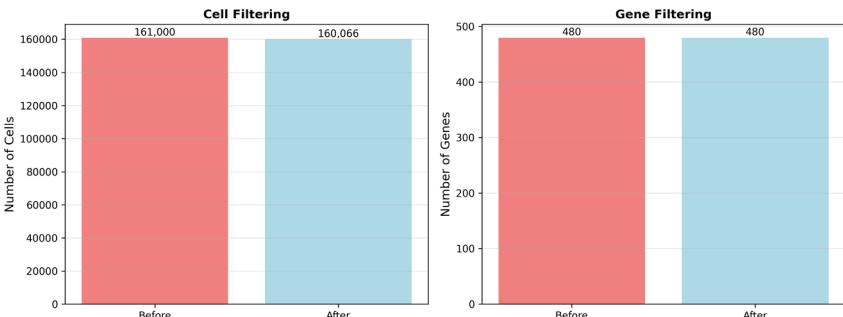


a Filtering Statistics



Domain	Primary Functional Domain Inference	Key Supporting Evidence (Top Enriched GO Terms & Biological Rationale)
0	T-cell-rich adaptive immune niche	Top terms: T cell activation (GO:0042110), lymphocyte differentiation (GO:0030098), regulation of immune effector process (GO:0002697). Strong enrichment for CD3D, CD3E, CD247, IL2RA. Suggests organized, antigen-experienced T-cell microenvironment — potentially tertiary lymphoid structure (TLS)-like or tumor-infiltrating lymphocyte (TIL) aggregate.
1	B-cell-dominant lymphoid aggregate	Top terms: B cell activation (GO:0042113), humoral immune response (GO:0006959), antibody-mediated immunity (GO:0002443). High load of CD19, MS4A1 (CD20), CD79A, IGHM. Consistent with germinal center-adjacent or ectopic B-cell follicle — functionally distinct from cluster 0, emphasizing humoral over cytotoxic immunity.
3	Regulatory T-cell / immunosuppressive interface	Top terms: regulation of T cell activation (GO:0050870), negative regulation of immune response (GO:0045089), T cell tolerance induction (GO:0001819). Co-enrichment of FOXP3, CTLA4, TGFBI, IL10RA. Suggests an immune-modulatory boundary zone — possibly tumor-stroma interface where suppression limits anti-tumor immunity.

f Validation Anchors in Available Data

- `sca_annotation_statistics.csv`: Confirms high Z-scores for T-cell and Treg markers — enabling confident interfacial spot classification.
- `giotto_cpdb_significant.csv`: Contains significant CTLA4-CD80, LAG3-HLA-DRA, TIGIT-PVRL2 (CD112), and TGFBI-TGFB2 interactions — supporting nanoscale polarization hypotheses.
- `enrichment_volcano.png` & `enrichment_results.csv`: Show strong, opposing enrichment for IFN- γ vs TGF- β pathways — validating functional antagonism at the interface.
- `annotation_umap.png`: Likely reveals tight clustering of T cells and Tregs in UMAP space — consistent with strong intercellular adhesion and synapse formation.

